

Logistic Regression Example Code

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Credit Card Default Data

A simulated data set containing information on ten thousand customers. The aim here is to predict which customers will default on their credit card debt.

default A binary variable indicating whether the customer defaulted on their debt ("No" or "Yes")

student A binary variable indicating whether a customer is a student ("No" or "Yes")

balance The average balance that the customer has remaining on their credit card after making their monthly payment

income Income of customer

Loading the data

```
# load package that contains the dataset
library(ISLR)
# get dataset from package
data(Default)

options(scipen=999) # this just turns off scientific notation

# look at structure of our binary variables (we want them to be in factor form - they are)
str(Default$default)
str(Default$student)
```

Fitting model and getting confidence intervals

```
# fit model
fit <- glm(default~student+balance+income, data = Default, family = "binomial")
summary(fit)

# get confidence intervals
confint(fit)
```

Prediction

```
# create newdata that we want to predict for
newdat <- data.frame(student="No",balance=2020,income=48500)

# get predictions on the logodds scale
pred_logodds <- predict(fit, newdata = newdat, type = "link")
pred_logodds

# get predicted odds
exp(pred_logodds)

# get predictions on the probability scale
pred_prob <- predict(fit, newdata = newdat, type = "response")
pred_prob
```

Confusion matrix

```
# calculate predicted probabilities for the dataset that we used to fit the model
probs <- predict(fit, type = "response")

# create a variable the same length as probs to store our classifications (just populate it with Nos)
classif <- rep("No",length(probs))
# replace the No in classif with a Yes if the probability in probs is greater than 0.3
classif[probs>0.3] <- "Yes"

# generate confusion matrix
table(classif, Default$default)
```